

Gurugram University Scheme of Studies and Examination
Bachelor of Technology (Electronics & Communication Engineering)

SEMESTER V

S. No	Category	Course Code	Course Title	Hours per week			Credits	Marks for Sessional	Marks for End Term Examination	Total
				L	T	P				
1	PCC		Computer Networks	3	0	0	3	30	70	100
2	PCC		Digital System Design	3	0	0	3	30	70	100
3	PCC		Digital Signal Processing	3	0	0	3	30	70	100
4	PCC		Wireless & Satellite Communication	3	0	0	3	30	70	100
5	PEC		Program Elective - I	3	0	0	3	30	70	100
6	OEC		Open Elective - I	3	0	0	3	30	70	100
7	LC		Computer Networks Lab	0	0	2	1	50	50	100
8	LC		Digital Signal Processing Lab	0	0	2	1	50	50	100
10	LC		Digital System Design Lab	0	0	2	1	50	50	100
11	LC		Wireless & Satellite Communication lab	0	0	2	1	50	50	100
12	MC		Practical Training-I	0	0	2	-	30	70	100*
Total				28			22	380	620	1000

NOTE:

1. Choose any one from Professional Elective Course-I
2. Choose any one from Open Elective Course-I
3. ***Practical Training-I:** The examination of the regular students will be conducted by the concerned college/Institute internally. Each student will be required to score a minimum of 40% marks to qualify in the paper. The marks will not be included in determining the percentage of marks obtained for the award of a degree.

PROFESSIONAL ELECTIVE- I (Semester-V)

Sr. No	Code	Subject	Credit
1.		Information Theory and Coding	3
2.		Antennas and Propagation	3
3.		Bio-Medical Electronics	3
4.		Introduction to MEMS	3
5.		Mobile applications development	3
6.		Analog Integrated Circuits	3

COMPUTER NETWORKS

Course Code						
Category	Professional Core Courses					
Course title	Computer Network					
Scheme	L	T	P	Credits	Semester : V	
	3	1	0	3		
Class Work	30 Marks					
Exam	70 Marks					
Total	100 Marks					
Duration of Exam	03Hrs					

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

COURSE OBJECTIVE:

1. To develop an understanding of modern network architectures from a design and Performance perspective.
2. To introduce the student to the major concepts involved in wide-area networks (WANs), local area networks (LANs), and Wireless LANs (WLANs).
3. To provide an opportunity to do Network programming.
4. To provide WLAN measurement ideas.

UNIT – I

Introduction: Data communication, Components, Data Representation, Simplex, Half Duplex, and Full Duplex Transmission, Modulation, Multiplexing, Computer networks, distributed processing, Internet, Topologies, Packet and circuit switching, connectionless and connection-oriented services.

Network Models: OSI model and TCP/IP Model

Physical Layer – LAN: Ethernet.

UNIT – II

Data Link Layer and Medium Access Sub Layer: MAC Addressing, Framing, Stop and Wait, Go back – N ARQ, Selective Repeat ARQ, Sliding Window Protocol.

Medium Access Control: Random access, Controlled Access, and channelization protocols.

Network Layer: Logical addressing, classful and classless addressing, subnetting, Ipv4, ICMPv4, ARP, RARP and BOOTP, Ipv6, Ipv6 addressing.

UNIT – III

Network Devices: Repeater, hub, switch, router, and gateway.

Routing Algorithms: introduction to routing, Shortest Path Algorithm, Flooding, Hierarchical Routing, Link State, and Distance Vector Routing

Transport Layer: Process to Process Communication, User Datagram Protocol (UDP), Transmission Control Protocol (TCP), TCP connection management.

UNIT – IV

Congestion Control, Quality of Service, QoS Improving techniques.

Application Layer: Domain Name Space (DNS), EMAIL, File Transfer Protocol (FTP), HTTP, SNMP

Network Security: Firewalls, security goals, types of attack, symmetric and asymmetric key ciphers.

COURSE OUTCOMES: At the end of this course, students will demonstrate the ability to

1. Explain the functions of the different layers of the OSI Protocol.
2. Draw the functional block diagram of wide-area networks (WANs), local area networks (LANs), and Wireless LANs (WLANs) and describe the function of each.
3. Identify and connect various connecting components of a computer network.
4. Configure DNS DDNS, TELNET, EMAIL, File Transfer Protocol (FTP), WWW, HTTP, SNMP, Bluetooth, and Firewalls using open-source available software and tools.
5. outline various models, topologies and devices of Computer Networks.
6. Design engineering solutions to complex problems utilizing a systems approach.

TEXT AND REFERENCE BOOKS:

1. Data Communication and Networking, 4th Edition, Behrouz A. Forouzan, McGraw-Hill.
2. Data and Computer Communication, 8th Edition, William Stallings, Pearson Prentice Hall India.
3. Computer Networks, latest Edition, Andrew S. Tanenbaum, Pearson New International Edition.
4. Internetworking with TCP/IP, Volume 1, latest Edition Douglas Comer, Prentice Hall of India.
5. TCP/IP Illustrated, Volume 1, W. Richard Stevens, Addison-Wesley, United States of America.

DIGITAL SYSTEM DESIGN

Course Code					
Category	Professional Core Courses				
Course title	Digital System Design				
Scheme	L	T	P	Credits	Semester : V
	3	1	0	3	
Class Work	30 Marks				
Exam	70 Marks				
Total	100 Marks				
Duration of Exam	03Hrs				

NOTE: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objective: The objectives of this course are as under:

1. To know the basic language features of Verilog HDL and the role of HDL in digital logic design.
2. To know the behavioural modeling of combinational and simple sequential circuits.
3. To know the behavioral modeling of algorithmic state machines.
4. To know the synthesis of combinational and sequential descriptions.
5. To know the architectural features of programmable logic devices.

UNIT I

Hardware modeling with the Verilog HDL: Encapsulation, modeling primitives, Types of Modelling. Logic system, Data types and operators. Behavioural descriptions in Verilog HDL. Styles for Synthesis of combinational logic and sequential logic. HDL based Synthesis – Technology Independent design

UNIT II

System Verilog standards, Key System Verilog enhancements for hardware design. Advantages of System Verilog over Verilog, Data Types: Verilog data types, System Verilog data types, 2 - State Data types, Bit, byte, shortint, int, longint. 4 - State data types. Logic, Enumerated data types, User Defined data types, Struct data types, Strings, Packages, Type Conversion: Dynamic casting, Static Casting, Memories: Arrays, Dynamic Arrays, Multidimensional Arrays, Packed Arrays, Associative Arrays, Queues, Array Methods, Tasks and Functions: Verilog Tasks and Functions

UNIT III

Verilog interface signals - Limitations of Verilog interface signals, SystemVerilog interfaces, SystemVerilog port connections, Interface instantiation. Interfaces Arguments, Interface Modports, Interface References, Tasks and functions in interface, Verilog Event Scheduler, SystemVerilog Event Scheduler, Clocking Block, Input and Output Skews, Typical Testbench Environment, Verification plan

UNIT IV

Random Variables - rand and randc, Randomize() Method - Pre/Post Randomize() methods, Constraints in the class, Rand mode and constraint mode, Constraint and Inheritance, Constraint Overriding, Set Membership, Distribution Constraints, Conditional Constraints - .implication (->), if/else, Inline Constraints

COURSE OUTCOMES: After successful completion of the course, the students are able to

1. Demonstate knowledge on HDL design flow,digital circuits design ,switch de-bouncing, metastability, memory devices applications
2. Can synthesis of combinational and sequential descriptions.
3. Design and develop the combinational and sequential circuits using behavioral modelling
4. Solving algorithmic state machines using hardware description language
5. Analyze the process of synthesizing the combinational and sequential descriptions
6. Memorizing the advantages of programmable logic devices and their description in Verilog

Reference Book

1. Samir Palnitkar “Verilog HDL A Guide to Digital Design Synthesis , “ 2nd Edition, Pearson Education 2006.
2. Ashenden - Digital design, Elsevier
3. IEEE Standard VHDL Language Reference Manual latest edition
4. Digital Design and Modelling with VHDL and Synthesis : KC Chang; IEEE Computer Society Press.
5. "A VHDL Primer" : Bhasker; Prentice Hall latest edition.
6. “Digital System Design using VHDL” : Charles. H. Roth ; PWS latest edition
7. "VHDL-Analysis & Modelling of Digital Systems” : Navabi Z; McGraw Hill.
8. VHDL-IV Edition: Perry; TMH latest edition
9. “Introduction to Digital Systems” : Ercegovac. Lang & Moreno; John Wiley latest edition
10. Fundamentals of Digital Logic with VHDL Design : Brown and Vranesic; TMH latest edition
11. Modern Digital Electronics- III Edition: R.P Jain; TMH latest edition.
12. Grout - Digital system Design using FPGA & CPLD 'S, Elsevier.

DIGITAL SIGNAL PROCESSING

Course Code						
Category	Professional Core Courses					
Course title	Digital Signal Processing					
Scheme	L	T	P	Credits	Semester : V	
	3	1	0	3		
Class Work	30 Marks					
Exam	70 Marks					
Total	100 Marks					
Duration of Exam	3Hrs					

NOTE: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objective: The objectives of this course are as under:

1. To describe signals mathematically and understand how to perform mathematical operations on signals.
2. Get familiarized with various structures of IIR and FIR systems.
3. To discuss word length issues , multi rate signal processing and application.
4. Design and realize various digital filters for digital signal processing.

UNIT I

Discrete time signals: Sequences; representation of signals on orthogonal basis; Sampling and reconstruction of signals; Discrete systems attributes, Z-Transform, Analysis of LSI systems, frequency Analysis, Inverse Systems,

UNIT II

Introduction to DFT: Efficient computation of DFT Properties of DFT – FFT algorithms – Radix-2 and Radix-4 FFT algorithms – Decimation in Time – Decimation in Frequency algorithms – Use of FFT algorithms in Linear Filtering and correlation.

UNIT III

Structure of IIR: System Design of Discrete time IIR filter from continuous time filter – IIR filter design by Impulse Invariance. Bilinear transformation – Approximation derivatives – Design of IIR filter in the Frequency domain. : Symmetric & Anti-symmetric FIR filters: Linear phase filter – Windowing techniques – rectangular, triangular, Blackman and Kaiser windows – Frequency sampling techniques – Structure for FIR systems.

UNIT IV

Finite word length effects in FIR and IIR digital filters: Quantization, round off errors and overflow errors. Multi rate digital signal processing: Concepts, design of practical sampling rate converters, Decimators, interpolators. Polyphase decompositions. Application of DSP – Model of Speech Wave Form – Vocoder.

COURSE OUTCOMES: After successful completion of the course, the students are able to

1. Interpret and analyze discrete time signals.
2. Compute Z transform.
3. Compute Discrete Fourier Transform.
4. Appreciate the importance of Fast Fourier Transform.
5. Design IIR and FIR filters.
6. Apply signal processing algorithms for real time applications.

Text Books

1. Digital Signal Processing A. Vallavaraj, C. Gnanapriya, and S. Salivahanan\
2. S.K. Mitra, Digital Signal Processing: A computer based approach.TMH
3. Oppenheim A V, Willsky A S and Young I T, "Signal & Systems", Prentice Hall, (1983).
4. Ifeachor and Jervis, "Digital Signal Processing", Pearson Education India.
5. DeFatta D J, Lucas J G and Hodgkiss W S, "Digital Signal Processing", J Wiley and Sons, Singapore, 1988

WIRELESS & SATELLITE COMMUNICATION

Course Code					
Category	Professional Core Courses				
Course title	Wireless & Satellite Communication				
Scheme	L	T	P	Credits	Semester : V
	3	1	0	3	
Class Work	30 Marks				
Exam	70 Marks				
Total	100 Marks				
Duration of Exam	3Hrs				

NOTE: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objective: The objectives of this course are as under:

1. To introduce the fundamentals of satellite communication .
2. To introduce roles of various sub-systems of a satellite system.
3. The course introduces the concept of Modulation and Multiple Access Schemes.
4. Study the satellite link design.
5. Study the satellite orbits.

UNIT I

Introduction to Satellite Communication: Principles and architecture of satellite Communication, Brief history of Satellite systems, advantages, disadvantages, applications and frequency bands used for satellite communication. IRNSS-NAVIC: Navigation with Indian Constellation

Orbital Mechanics: Orbital equations, Kepler's laws, Apogee and Perigee for an elliptical orbit, evaluation of velocity, orbital period, angular velocity etc of a satellite, concepts of Solar day and Sidereal day.

UNIT II

Satellite sub-systems: Study of Architecture and Roles of various sub-systems of a satellite system such as Telemetry, tracking, command and monitoring (TTC & M), Attitude and orbit control system (AOCS), Communication sub-system, power sub-systems etc

UNIT III

Typical Phenomena in Satellite Communication: Solar Eclipse on satellite, its effects, remedies for Eclipse, Sun Transit Outage phenomena, its effects and remedies, Doppler frequency shift phenomena and expression for Doppler shift.

Satellite link budget : Flux density and received signal power equations, Calculation of System noise temperature for satellite receiver, noise power calculation, Drafting of satellite link budget and C/N ratio calculations in clear air and rainy conditions.

UNIT IV

Multiple Access Techniques For Wireless Communication: Introduction to Multiple Access, FDMA, TDMA, Spread Spectrum multiple Access, space division multiple access, packet ratio, capacity of a cellular systems.

Wireless Networking: Difference between wireless and fixed telephone networks, development of wireless networks, fixed network transmission hierarchy, traffic routing in wireless networks, wireless data services, common channel signaling, ISDN (Integrated Services digital Networks), advanced intelligent networks.

Course Outcomes: At the end of this course students will demonstrate the ability to

1. Understand the sub-systems of satellite communication systems and ground stations.
2. Understand the signal power calculation and issues in communication satellite tracking.
3. Compute parameters of orbital motions and understand communication with non-geosynchronous satellite
4. Understand different modulation types and interfacing the modems in satellite receivers.
5. Understand various multiple access technique.
6. Applications and frequency bands used for satellite communication.

Text/Reference Books:

1. Timothy Pratt and Jeremy Allnutt: "Satellite Communications": Ed 3, 2021. Wiley India.
2. Dennis Roddy: Satellite Communication: 4th Edition, McGraw Hill, 2001
3. Varsha Agrawal, Anil K. Maini, "Satellite Communications" Wiley India 2010.
4. Tri T. Ha: Digital Satellite Communications: Tata McGraw Hill.

DIGITAL SYSTEM DESIGN LABORATORY

Course Code						
Category	Laboratory Courses					
Course title	Digital System Design Laboratory					
Scheme	L	T	P	Credits	Semester: V	
	0	0	2	1		
Class Work	50 Marks					
Exam	50 Marks					
Total	100 Marks					
Duration of Exam	2Hrs					

Note:

1. At least 10 experiments are to be performed by students in the semester
2. At least 7 experiments should be performed from the list, remaining three experiments may either be performed from the above list or designed and set by the concerned institution as per the scope of the syllabus.
3. Group of students for practical should be 15 to 20 in number.

List of Experiments

Combinational & Sequential Design Exercises using HDL

1. Design a Half-Adder & Full Adder.
2. Design a Half Subtractor, & Full Subtractor
3. Design a parity generator
4. Design a 4 Bit comparator
5. Design a RS & JK Flip flop
6. Design a 4: 1 Multiplexer
7. Design a 4 Bit Up / Down Counter with Loadable Count
8. Design a 3: 8 decoder
9. Design a 8 bit shift register
10. Design a arithmetic unit

FPGA (Spartan 3) & CPLD

11. Implement ADC & DAC interface with FPGA
12. Implement a serial communication interface with FPGA
13. Implement a Telephone keypad interface with FPGA
14. Implement a VGA interface with FPGA
15. Implement a PS2 keypad interface with FPGA
16. Implement a 4-digit seven segment display

Lab outcomes:

1. Identify the various IC fabrication methods.
2. Express the Layout of simple MOS circuit using Lambda based design rules.
3. Apply the Lambda based design rules for subsystem design
4. Differentiate various FPGA architectures.
5. Design an application using Verilog HDL.
6. Concepts of modeling a digital system using Hardware Description Language.

DIGITAL SIGNAL PROCESSING LABORATORY

Course Code						
Category	Laboratory Courses					
Course title	Digital Signal Processing Laboratory					
Scheme	L	T	P	Credits	Semester: V	
	0	0	2	1		
Class Work	50 Marks					
Exam	50 Marks					
Total	100 Marks					
Duration of Exam	2Hrs					

Notes:

1. At least 10 experiments are to be performed by students in the semester.
2. At least 7 experiments should be performed from the list, remaining three experiments may either be performed from the above list or designed and set by the concerned institution as per the scope of the syllabus.
3. Group of students for practical should be 15 to 20 in number.

List of Experiments

11. Introduction to MATLAB.
12. Represent basic signals (unit step, unit impulse, ramp, exponential, sine and cosine)
2. To develop program for Z-Transform in MATLAB
3. To develop program for Convolution of sequences in MATLAB
4. To develop program for Correlation of sequences in MATLAB
5. To develop program for DFT & IDFT of two sequences
6. To develop program for FFT of two Sequences
7. To develop program for Circular Convolution
8. To design analog filter (low-pass, high pass, band-pass, band-stop).
9. To design digital IIR filters (low-pass, high pass, band-pass, band-stop).
10. To develop program for Interpolation and Decimation of sequences
11. To design FIR filters using windows technique.
12. Detection of Signals buried in Noise
13. Effect of noise on signals in MATLAB

Lab Outcomes:

At the end of this lab, students will be able to

1. Interpret and analyze discrete time signals.
2. Compute Z transform.
3. Compute Discrete Fourier Transform.
4. Appreciate the importance of Fast Fourier Transform.
5. Design IIR and FIR filters.
6. Apply signal processing algorithms for real time applications.

COMPUTER NETWORKS LAB

Course code					
Category	Laboratory Courses				
Course title	Computer Networks Lab				
Scheme and Credits	L	T	P	Credits	Semester IV
	0	0	2	2	
Classwork	50 Marks				
Exam	50 Marks				
Total	100 Marks				
Duration of Exam	02 Hours				

COURSE OBJECTIVES:

1. To understand the functionalities of various layers of OSI model.
2. Understand fundamental underlying principles of computer networking.

CONTENTS:

1. Study of Socket Programming and Client – Server model
2. Write a code simulating ARP /RARP protocols.
3. Write a code simulating PING and TRACEROUTE commands
4. Create a socket for HTTP for web page upload and download.
5. Write a program to implement RPC (Remote Procedure Call)
6. Implementation of Sub netting .
7. Applications using TCP Sockets like a. Echo client and echo server b. Chat c. File Transfer
8. Applications using TCP and UDP Sockets like. DNS e. SNMP f. File Transfer
9. Study of Network simulator (NS).and Simulation of Congestion Control Algorithms using NS
10. Perform a case study about the different routing algorithms to select the network path with its optimum and economical during data transfer. i. Link State routing ii. Flooding iii. Distance vector
11. To learn handling and configuration of networking hardware like RJ, 45 connector, CAT,6 cable, crimping tool, etc.
12. Configuration of router, hub, switch etc. (using real devices or simulators)
13. Running and using services/commands like ping, traceroute, nslookuparp, telnet, ftp, etc.
14. Network packet analysis using tools like Wireshark, tcpdump, etc.

COURSE OUTCOMES: At the end of course , the student will be able to :

1. Develop Client , Server architectures and prototypes by the means of correct standards and technology
2. .Analyze data flow between peer to peer in an IP network using Application, Transport and Network Layer Protocols.
3. Analyse & Implement various framing methods of Data Link Layer.
4. Demonstrate basic configuration of switches and routers.
5. Analyse & Implement various Error and flow control techniques.
6. Implement network routing and addressing techniques.

WIRELESS & SATELLITE COMMUNICATION LABORATORY

Course Code						
Category	Laboratory Courses					
Course title	Wireless & Satellite Communication Laboratory					
Scheme	L	T	P	Credits	Semester: V	
	0	0	2	1		
Class Work	50 Marks					
Exam	50 Marks					
Total	100 Marks					
Duration of Exam	2Hrs					

Notes:

- (i) At least 10 experiments are to be performed by students in the semester.
- (ii) At least 7 experiments should be performed from the list, remaining three experiments may either be performed from the above list or designed and set by the concerned institution as per the scope of the syllabus. Group of students for practical should be 15 to 20 in number.

LIST OF EXPERIMENTS:

1. To set up a satellite communication link & study of change in uplink & downlink frequency.
2. To Study Transmission of Audio & Video Signals & Data communication over satellite link.
3. To Study Transmission of telemetry data like temperature & light intensity over satellite link
4. To measure the propagation delay of signal in a Satellite communication Link.
5. To study different GPS data like longitude, latitude & different types of dilute of precision using GPS receiver..
6. To study selection of various PN codes like Gold, Barker & MLS in CDMA technology .
7. To study generation (spreading) & demodulation (Despreading) of of DSSS modulated signal
8. To study Voice communication over DSSS.
9. To study Minimum shift keying modulation & de modulation.
10. To study radiation pattern & calculate beam width for Yagi uda & Folded dipole antenna.
11. To study radiation pattern & calculate beam width for Circular & Triangular Patch Antenna.
12. to study FHSS Modulation & demodulation & transfer of numeric data.

Lab Outcomes: After completion of this course, students will be able to :

7. Describe the basic components of satellite communication system.
8. Understand transmission of audio & video signal, telemetry data, GPS data and modulation techniques.
9. Demonstrate radiation pattern and calculate beam width for different antennas.
10. Analyze the propagation delay of signals in satellite communication links.
11. Understand CDMA, spread spectrum techniques.

PRACTICAL TRAINING-I

Course Code						
Category	PT					
Course title	Practical Training-I					
Scheme	L	T	P	Credits	Semester: V	
	2	0	0	0		
Class Work	30					
Exam	70					
Total	100					
Duration of Exam	3Hrs					

Note: The examination of the regular students will be conducted by the concerned college/Institute internally. Each student will be required to score a minimum of 40% marks to qualify in the paper. The marks will not be included in determining the percentage of marks obtained for the award of a degree.

The students are required to undergo practical training of duration not less than 1.5 months in a reputed organization or concerned institute. The students who wish to undergo practical training, the industry chosen for undergoing the training should be at least a private limited company. The students shall submit and present the midterm progress report at the institute. The presentation will be attended by a committee. alternately the teacher may visit the industry to get the feedback of the student.

The final Viva voice of the practical training will be conducted by an external examiner and one external examiner appointed by the institute. External examiner will be from the panel of examiners submitted by the concerned institute approved by the board of studies in engineering and technology. Assessment of industrial training will be based on seminar, viva-voice, report and certificate of practical training obtained by the student from the industry or institute.

PROFESSIONAL ELECTIVE – I

Information Theory and Coding

Course Code						
Category	Professional Elective Courses					
Course title	Information Theory and Coding					
Scheme	L	T	P	Credits	Semester: V	
	3	0	0	3		
Class Work	30 Marks					
Exam	70 Marks					
Total	100 Marks					
Duration of Exam	3Hrs					

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objective: The objectives of this course are as under:

1. Students study the basics of Information theory, techniques of coding and decoding.
2. Study to analyze and compare different coding and decoding schemes.
3. Solve numerical problems on channel capacity and coding.
4. Study broadcast channels for different coding schemes and also multiuser channel coding.

UNIT - I

Basics of information theory: Entropy for discrete ensembles; Shannon's noiseless coding theorem; Encoding of discrete sources. Markov sources, Shannon's noisy coding theorem and converse for discrete channels, Calculation of channel capacity and bounds for discrete channels, application to continuous channels.

UNIT - II

Techniques of coding and decoding: Channel Coding, Block and convolutional codes; majority logic decoding; Viterbi decoding algorithm, Coding gains and performance. Huffman codes and uniquely detectable codes; Cyclic codes, convolutional arithmetic codes.

UNIT - III

Network Information Theory: Overview of multiple access channel (MAC), Achievable result for MAC using successive decoding technique, Outer bound on the capacity region of MAC and its capacity analysis, Gaussian MAC and its capacity analysis.

Introduction to broadcast channel: Superposition coding scheme and its optimality for the degraded broadcast channel, Relation between the capacity region of Gaussian BC and MAC. Achievable rate for interference limited networks using conventional techniques such as time-sharing and treating interference as noise.

UNIT - IV

Introduction to channel coding for multi users: Introduction, Block codes for the binary adder channel, Trellis codes for the multiple access channel.

Course Outcomes: At the end of this course students will demonstrate the ability to

1. To Study and Derive equations for entropy mutual information and channel capacity for all types of channels.
2. To acquire the knowledge about Fourier series and Fourier transform signal analysis tool.
3. Design a digital communication system by selecting an appropriate error correcting codes for a particular application.
4. To learn about Probability of Random signal theory and process.
5. Formulate the basic equations of linear block codes and a cyclic code.
6. Compare the performance of digital communication system by evaluating the probability of error for different error correcting codes.

Text/Reference Books:

1. N. Abramson, Information and Coding, McGraw Hill, 1963.
2. M. Mansurpur, Introduction to Information Theory, McGraw Hill, 1987.
3. R.B. Ash, Information Theory, Prentice Hall, 1970.
4. Shu Lin and D.J. Costello Jr., Error Control Coding, Prentice Hall, 1983.
5. A. El Gamal and Y. H. Kim, Network Information Theory, Cambridge University Press, 2011

BIO-MEDICAL ELECTRONICS

Course Code						
Category	Professional Elective Courses					
Course title	Bio-medical Electronics					
Scheme	L	T	P	Credits	Semester: V	
	3	0	0	3		
Class Work	30 Marks					
Exam	70 Marks					
Total	100 Marks					
Duration of Exam	3Hrs					

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objective: The objectives of this course are as under:

1. To understand concept of electronic systems design in Bio- medical applications.
2. To understand the biological process.
3. To understand non electrical parameter measurements.
4. To understand various Bio Medical Measuring Instruments and therapeutic equipments.

UNIT-I

Physiology and Transducers

Brief introduction to human physiology: Cell and its structure; Resting and Action Potential; Nervous system: Functional organisation of the nervous system; Structure of nervous system, neurons; synapse; transmitters and neural communication; Cardiovascular system; respiratory system; Basic components of a biomedical system. Biomedical transducers: Transducers selection criteria; Piezoelectric; ultrasonic; displacement, velocity, force, acceleration, flow, temperature, potential, dissolved ions and gases; Temperature measurements; Fibre optic temperature sensors

UNIT-II

Electro – Physiological Measurements : Bio-electrodes and Biopotential amplifiers for ECG, EMG, EEG, etc.: Limb electrodes; floating electrodes; pregelled disposable electrodes; Micro, needle and surface electrodes; Preamplifiers, differential amplifiers, chopper amplifiers; Isolation amplifier. ECG; EEG; EMG; ERG; Lead systems and recording methods

UNIT-III

Non-Electrical Parameter Measurements

Measurement of blood temperature, pressure and flow; Cardiac output; Heart rate; Heart sound; Pulmonary function measurements; spirometer; Impedance plethysmography; Photo Plethysmography, Body Plethysmography

Unit-IV

Medical Imaging, Ultrasonic, X-ray and nuclear imaging: Radio graphic and fluoroscopic techniques; Computertomography; MRI; Ultrasonography, Assisting And Therapeutic Equipments, Prostheses and aids: pacemakers, defibrillators, heart-lung machine, artificial kidney, aids for the handicapped; Safety aspects: safety parameters of biomedical equipments

Course Outcomes:

At the end of this course, students will demonstrate the ability to:

1. Apply the concept of electronic systems design in Bio- medical applications.
2. Examine the practical limitations on the electronic components while handling bio- substances.
3. Evaluate and analyze the biological processes like other electronic processes.
4. Measure non electrical parameter.
5. Familiar the various Bio Medical Measuring Instruments and the therapeutic equipments.
6. Aware of electrical safety of medical equipments

Text/ reference books:

1. W.F. Ganong, Review of Medical Physiology, latest edition, Medical Publishers
2. J.G. Webster, ed., Medical Instrumentation, Houghton Mifflin, latest edition
3. A.M. Cook and J.G. Webster, eds., Therapeutic Medical Devices, Prentice-Hall, latest edition
4. R.S.Khander, Handbook of Biomedical Instrumentation, TATA Mc Graw-Hill, New Delhi, latest edition
5. Leslie Cromwell, —Biomedical Instrumentation and Measurement, Prentice Hall of India, New Delhi, latest edition

ANTENNAS AND PROPAGATION

Course code					
Category	Professional Elective Courses				
Course title	Antennas and Propagation				
Scheme and Credits	L 3	T 0	P 0	Credits 3	Semester :V
Classwork	30 Marks				
Exam	70 Marks				
Total	100 Marks				
Duration of Exam	03 Hours				

NOTE: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objectives: During the course, students will be made to learn to:

1. Understand the working principles of the Antenna.
2. Understand the types of Antenna and their propagation.
3. Understand limitations and application for different networks.

UNIT - I

Fundamental Concepts- Physical concept of radiation, Radiation pattern, near-and far-field regions, reciprocity, directivity and gain, effective aperture, polarization, input impedance, efficiency, Friis transmission equation, radiation integrals and auxiliary potential functions.

Radiation from Wires and Loops- Infinitesimal dipole, finite-length dipole, linear elements near conductors, dipoles for mobile communication, small circular loop.

UNIT-II

Aperture and Reflector Antennas- Huygens' principle, radiation from rectangular and circular apertures, design considerations, Babinet's principle, Radiation from sectoral and pyramidal horns, design concepts, prime-focus parabolic reflector and cassegrain antennas.

Broadband Antennas- Log-periodic and Yagi-Uda antennas, frequency independent antennas, broadcast antennas.

UNIT – III

Micro strip Antennas- Basic characteristics of micro strip antennas, feeding methods, methods of analysis, design of rectangular and circular patch antennas.

Antenna Arrays- Analysis of uniformly spaced arrays with uniform and non-uniform excitation amplitudes, extension to planar arrays, synthesis of antenna arrays using Schelkunoff polynomial method, Woodward-Lawson method.

UNIT - IV

Basic Concepts of Smart Antennas- Concept and benefits of smart antennas, Fixed weight beamforming basics, Adaptive beamforming. Different modes of Radio Wave propagation used in current practice.

Course outcomes: At the end of this course students will demonstrate the ability to

1. Understands Physical concept of radiation, Radiation pattern
2. Understand antenna characteristics for different applications.
3. Analyze and design different types of antennas.
4. Understands basic concept of smart Antennas.
5. Design antenna arrays and understand operation of smart antennas.
6. Investigate different modes of propagation and their suitability for wireless communication

Text/Reference Books:

1. J.D. Kraus, Antennas, McGraw Hill, 1988.
2. C.A. Balanis, Antenna Theory - Analysis and Design, John Wiley, 1982.
3. R.E. Collin, Antennas and Radio Wave Propagation, McGraw Hill, 1985.
4. R.C. Johnson and H. Jasik, Antenna Engineering Handbook, McGraw Hill, 1984.
5. I.J. Bahl and P. Bhartia, Micro Strip Antennas, Artech House, 1980.
6. R.K. Shevgaonkar, Electromagnetic Waves, Tata McGraw Hill, 2005
7. R.E. Crompton, Adaptive Antennas, John Wiley

Analog Integrated Circuit

Course Code						
Category	Professional Elective Courses					
Course title	Analog Integrated Circuit					
Scheme	L	T	P	Credits	Semester : V	
	3	1	0	3		
Class Work	30 Marks					
Exam	70 Marks					
Total	100 Marks					
Duration of Exam	03Hrs					

NOTE: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

Course Objective: The objectives of this course are as under:

1. To understand the functioning of OP-AMP and design OP-AMP based circuits
2. To design and analyze waveform Generators.
3. To design sinusoidal and non-sinusoidal oscillators
4. To understand the concept of filters and regulators.

UNIT I

IC OP-AMP applications: OP-AMP Fundamentals (brief review of differential amplifier, current mirror, active load, level shifter, output stage; ac and dc characteristics) Basic building blocks using OP-AMPS. Inverting/Non-inverting VCVS, Integrators, Differentiators, C CVS and VCCS, Instrumentation Amplifiers.

UNIT II

Waveform Generator: Square wave generators: 555Timer, Crystal controlled Oscillator Ramp Generator: Triangle generator, Sawtooth generator Sine wave generator: Requirement for sinusoidal oscillations, Wien-bridge and twin-T oscillators. Function Generators: Multi op-amp function generators, IC function generators Digitally controlled frequency synthesizer: PLL Fundamentals, PLL synthesizer, Totally digital synthesizer.

UNIT III

Active Filters: Introduction to filtering: Frequency response, Characteristics and terminology, Active versus passive filters Low pass filter: First order low pass active filter, second order active filter model, second order low pass filter characteristics, Sallen-Key unity gain filter, Sallen-Key equal component filter, Higher order filters. High pass active filter. Band pass filter: single op-amp band pass filter, multistage band pass filter State variable filter.

UNIT IV

Non-linear Circuits: Logarithmic Amplifiers, Log/Antilog Modules, Precision Rectifier, Peak Detector, Sample and Hold Circuits. OP-AMP as Comparator, Schmitt Trigger, Square and Triangular Wave Generator, Monostable Multivibrator. IC Analog Multiplier applications OTA
Voltage Regulators: OP-AMP Regulators, IC Regulators, Fixed Voltage Regulators (78/79, XX), SMPS.

Course Outcomes: At the end of this course, students will demonstrate the ability to:

1. Design, analysis, simulations, and testing of analog circuits,
2. Analog electronics, with focus on integrated circuit design
3. DC biasing, op-amp Cadence for Simulation Labs and Homework problems.
4. Understand filters and related terminology.
5. Concept of voltage regulators, SMPS
6. Different form of waves and how they generated.

Text Book:

1. Sedra and Smith, Microelectronic Circuits”, Oxford University press, 5th Edition, 2005.
2. J. Michael Jacob, Applications and design with Analog Integrated Circuits”, PHI, 2nd Edition, 2004.

Reference Book :

B.P. Singh and Rekha Singh, Electronic Devices an Integrated Circuits; Pearson Education, 1st Edition 2006.

MOBILE APPLICATIONS DEVELOPMENT

Course code					
Category	Professional Elective Courses				
Course title	Mobile applications development				
Scheme and Credits	L	T	P	Credits	Semester: V
	3	0	0	3	
Classwork	30 Marks				
Exam	70 Marks				
Total	100 Marks				
Duration of Exam	03 Hours				

Note: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

COURSE OBJECTIVE:

1. Introduce the students with the various “Next Generation Technologies” in the area of mobile computing
2. Assist students understand the various Mobile operating Systems
3. Explore the findings using Android Technologies

UNIT - I

Introduction: Mobile operating system, Operating system structure, Constraints and Restrictions, Hardware configuration with mobile operating system, Features: Multitasking Scheduling, Memory Allocation, File System Interface, Keypad Interface, I/O Interface, Protection and Security, Multimedia features

UNIT - II

Introduction to Mobile development IDE's, Introduction to Worklight basics, Optimization, pages and fragments , Writing a basic program- in Worklight Studio, Client technologies, Client side debugging, Creating adapters, Invoking adapters from Worklight Client application, Common Controls, Using Java in adapters, Programming exercise with Skins, Understanding Apache Cordova.

UNIT - III

Understanding Apple iOS development, Android development, Shell Development, Creating Java ME application, Exploring the Worklight Server, Working with UI frameworks, Authentication, Push notification, SMS Notifications, Globalization.

UNIT - IV

Android: Introduction to Android, Architecture, memory management, communication protocols, application development methods, deployment. **iOS:** Introduction to iOS, Architecture, memory management, communication protocols, application development

methods, deployment

COURSE OUTCOMES: At the end of this course, students will demonstrate the ability to

1. Explain the principles and theories of mobile computing technologies.
2. Describe infrastructures and technologies of mobile computing technologies.
3. List applications in different domains that mobile computing offers to the public, employees, and businesses.
4. Describe the possible future of mobile computing technologies and applications.
5. Effectively communicate course work through written and oral presentations

TEXT AND REFERENCE BOOKS:

1. Anubhav Pradhan, Anil V Deshpande, “ Mobile Apps Development” Edition:
2. Jeff McWherter, Scott Gowell “Professional Mobile Application Development”, John Wiley & Sons, 2012.
3. Barry Burd, “Android Application Development All in one for Dummies”, Edition: I
4. Teach Yourself Android Application Development In 24 Hours, Edition: I, Publication: SAMS
5. Neal Goldstein, Tony Bove, “iPhone Application Development All-In-One For Dummies”, John Wiley & Sons
6. Henry Lee, Eugene Chuvyrov, “Beginning Windows Phone App Development”, Apress, latest edition.
7. Jochen Schiller, “Mobile Communications”, Addison-Wesley, latest edition
8. Stojmenovic and Cacute, “Handbook of Wireless Networks and Mobile Computing”, Wiley, 2002, ISBN 0471419028.

INTRODUCTION TO MEMS

Course code					
Category	Professional Elective Courses				
Course title	Introduction to MEMS				
Scheme and Credits	L	T	P	Credits	Semester :V
	3	0	0	3	
Classwork	30 Marks				
Exam	70 Marks				
Total	100 Marks				
Duration of Exam	03 Hours				

NOTE: The examiner will set nine questions in total. Question one will be compulsory. Question one will have seven parts of 2 marks each from all units, and the remaining eight questions of 14 marks each to be set by taking two questions from each unit. The students have to attempt five questions in total, the first being compulsory and selecting one from each unit.

COURSE OBJECTIVES: The objectives of this course are as under:

1. To understand concept of Microsystems and Microelectronics.
2. To understand the MEMS fabrication modules.
3. To understand Scaling effect,sensors.

UNIT - I

Introduction and Historical Background , Introduction Microsystems vs. MEMS, Microsystems and Microelectronics, the Multidisciplinary Nature of Microsystems design and manufacture, Application of MEMS in various industries. MEMS and Miniaturization: Scaling laws in miniaturization: Scaling Effects. Micro/ Nano Sensors, Actuators and Systems overview: Case studies.

UNIT - II

Review of Basic MEMS fabrication modules: Oxidation, Deposition Techniques, Lithography (LIGA), Etching. Micromachining: Surface Micromachining, sacrificial layer processes, Stiction; Bulk Micromachining, Isotropic Etching and Anisotropic Etching, Wafer Bonding.

UNIT - III

Mechanics of solids in MEMS/NEMS: Stresses, Strain, Hookes's law, Poisson effect, Linear Thermal Expansion, Bending; Energy methods,

UNIT - IV

Overview of Finite Element Method, Modeling of Coupled Electromechanical Systems. Electrostatics, coupled electro mechanics.

Course Outcomes: At the end of this course students will demonstrate the ability to

1. Understand the multidisciplinary aspects of MEMS and NEMS.
2. MEMS and NEMS applications ,also learn sensors and actuators
2. Understand the methods of fabrication and modeling methods.
3. Appreciate the underlying working principles of MEMS and NEMS devices
4. Design and model these devices.
5. Understand mechanics of solid in MEMS and NEMS

Text/Reference Book:

1. G. K. Ananthasuresh, K. J. Vinoy, S. Gopalkrishnan K. N. Bhat, V. K. Aatre, Micro and Smart Systems, Wiley India, 2012.
2. S. E. Lyshevski, Nano-and Micro-Electromechanical systems: Fundamentals of Nano-and Microengineering (Vol. 8). CRC press, (2005).
3. S. D. Senturia, Microsystem Design, Kluwer Academic Publishers, 2001.
4. . M. Madou, Fundamentals of Microfabrication, CRC Press, 1997.
5. . G. Kovacs, Micromachined Transducers Sourcebook, McGraw-Hill, Boston, 1998.
6. . M.H. Bao, Micromechanical Transducers: Pressure sensors, accelerometers, and gyroscopes, Elsevier, New York, 2000.